

hacking or phishing attempt. Also, all the containers share the same Linux kernel; so if there's any vulnerability in the kernel itself, it will affect all the containers attached to it. This is the reason why some people consider virtual machines far more secure than Linux containers. Although VMs are not totally secure due to the presence of the hypervisor, the latter is still less vulnerable due to its limited functionalities. A lot of progress has been made in making these containers safe and secure. Docker and other management systems these days have made it mandatory for their administrators to mark container images to avoid deployment of untrusted containers.

Here are some of the ways to make your containers more secure.

Updating the kernel: One of the most vulnerable components in container management is the kernel, since it is shared among all the containers. So special care should be taken to maintain the kernel at its latest update whenever it's available. But updating a kernel is a two-step process. The first step is to update the code of the kernel and the second is to reboot your kernel. Software like Ksplice updates your kernel without rebooting it.

Access controls: The deep understanding of two controls — discretionary access control (DAC) and mandatory access control (MAC) — can also add to the safety of the containers. As the name suggests, DAC comprises policies that can be modified depending upon the user running the process. One such example of DAC is file system permissions. For example, if the owner of the file has set the mode of the file to 0600, then that file will not be available for read or write purposes to other users. DAC provides very good safety cover to the containers and its contents, since it is easy to use. On the other hand, MAC provides the mandatory policies which can only be overridden after they have been changed or disabled entirely. These controls are difficult to troubleshoot since they are not visible on the system.

Securing system calls: Another way of securing container systems is to limit the system calls or syscalls, as they are known, so that the unwanted calls are not made by the processes. So, whenever a call is made by a process that is not certified, that call will be killed and the security of the system will be maintained.

Advantages and disadvantages of using containers

Advantages:

- Running different servers on one machine minimises the hardware resources used and the power consumption.
- Reduces time to market for the product since less time is required in development, deployment and testing of services.
- Containers provide great opportunities for DevOps and CI/CD.
- Space occupied by container based virtualisation is much less than that occupied by virtual machines.

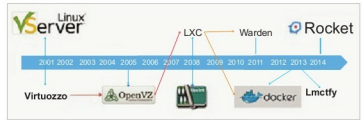


Figure 2: History of containers depicting the evolution from one system to another. *Image credit: kivenlau*

- Starting containers requires only a few seconds; so in data centres, they can be readily used in case of higher loads.

Disadvantages:

- Security is the main bottleneck in implementing and using container technology.
- Efficient networking is also required to connect containers deployed in isolated locations.
- Containerisation leads to fewer opportunities with regard to OS installation as compared to virtual machines.

The road ahead

The future of Linux containers looks very promising, since more and more organisations are moving towards them and away from virtual machines, due to the lower cost and greater flexibility. Tech giants like Google and Microsoft have also contributed to the container space, and used container technology to provide support for their products.

The Host cluster is also an upcoming trend in the IT world. Still in its initial stages, it should become more stable and robust in the coming years. Host cluster technology requires different containers to be run on different hosts to avoid single points of failure. There are various such open source projects like Apache Mesos, Kubernetes, Docker Swarm, etc.

Containers are also being used in cloud computing to provide cloud based services. Container based infrastructure is a fast evolving cloud technology because it is light in weight and high in performance.

It can now safely be said that Linux containers will completely replace virtual machines pretty soon. 

References

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